

Week 2-3 Worksheet: Problem Formulation & Data Planning (March 22 - April 5)

OR 6205 Road Trip Project: Boston to Florida

Part A: Defining Your Two Models (Complete Together)

Your project requires two models with distinctive goals. For each model, work through the sections below.

MODEL 1

A1.1 Objective Statement

Write your objective in one clear sentence. Use the format: "Minimize/Maximize [what] for [whom]."

A1.2 Why This Objective Matters

Who benefits from optimizing this objective? Why would a decision-maker care about this?

A1.3 Decision Variables

What decisions need to be made? List each decision variable and explain what it represents in plain language.

Variable	What It Represents	Units	Possible Values
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A1.4 Constraints

What limitations or requirements must your solution satisfy? For each constraint, identify whether it comes from the problem itself (external) or from assumptions your team is making (internal).

Constraint Description	Mathematical Form (if known)	Limitations?	Why This Constraint Exists
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A1.5 Assumptions

What assumptions are you making to formulate this model? For each assumption, explain why you made this choice and what alternative you considered.

Assumption (what is important or not – what is considered or not)	Why We Made This Choice (to include or not)	Alternative We Considered	How This Assumption Affects Results
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MODEL 2

A2.1 Objective Statement

Write your objective in one clear sentence. Use the format: "Minimize/Maximize [what] for [whom]."

A2.2 Why This Objective Matters

Who benefits from optimizing this objective? Why would a decision-maker care about this?

A2.3 Decision Variables

What decisions need to be made? List each decision variable and explain what it represents in plain language.

Variable	What It Represents	Units	Possible Values
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A2.4 Constraints

What limitations or requirements must your solution satisfy?

Constraint Description	Mathematical Form (if known)	External or Internal?	Why This Constraint Exists
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A2.5 Assumptions

What assumptions are you making to formulate this model?

Assumption	Why We Made This Choice (to include or not)	Alternative We Considered	How This Assumption Affects Results
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Part B: Comparing Your Two Models (Complete Together)

B1. Key Differences

How do your two models differ? What trade-offs exist between them?

B2. Stakeholder Impact

Would different stakeholders prefer different models? Explain.

B3. What Your Models Cannot Capture

Every model simplifies reality. What important aspects of the road trip problem do your models leave out? Why did you choose to exclude these?

Part C: Data Planning (Complete Together)

C1. Data Inventory

For each model, what data do you need? Complete the table for all required data.

Data Needed For Which Model? Source How Reliable? What If Unavailable?

C2. Data Collection Plan

Who will collect which data? By when?

Data Team Member Responsible Deadline Status

C3. Data Limitations

What limitations exist in your data sources? How might these limitations affect your conclusions?

Part D: Model Selection Justification (Complete Together)

Your project checklist asks you to complete this sentence for each modeling choice: "We chose X instead of Y and Z because [problem/decision-maker specific reason] and this choice means [specific consequence]."

Practice this now for your major decisions.

Decision 1: Choice of Objective Functions

"We chose to optimize _____ and _____ instead of _____ because _____. This choice means _____."

Decision 2: Problem Structure

"We chose to model this as a [LP/Network/other] problem instead of _____ because _____. This choice means _____."

Decision 3: Key Assumption

"We chose to assume _____ instead of _____ because _____. This choice means _____."

Part E: Critical Analysis Checkpoint (Complete Together)

Your project checklist emphasizes considering diverse perspectives and unintended consequences. Address these questions honestly.

E1. Whose Perspective Might You Be Missing?

Your team has backgrounds and experiences. What perspectives on this problem might you not naturally consider? How could this affect your modeling choices?

E2. Unintended Consequences

If someone followed your model's recommendations exactly, what could go wrong? What unintended consequences might arise?

E3. Values Embedded in Your Model

Your project checklist asks, "What values are embedded in my choice of objective function?" What does your choice of objectives reveal about what you think matters most? Would everyone agree?

Part F: Questions for April 6 Meeting (Complete Together)

What questions or uncertainties do you want to discuss with Professor Drummond at your results meeting on April 6?

1. _____
2. _____
3. _____

Submission Checklist

- ☐ Both models are fully described (Parts A1 and A2)
- ☐ Model comparison completed (Part B)
- ☐ Data planning completed (Part C)
- ☐ Justification sentences drafted (Part D)
- ☐ Critical analysis completed (Part E)
- ☐ Questions prepared for April 6 meeting (Part F)
- ☐ Each team member has submitted their Individual Reflection separately on Canvas